

$$x\ddot{y} - 4\dot{x}$$

TECHNIQUES

nd polyn. sol., plot sol., plot deriv., value in u f2

```

c := ...
sol := dsolve(ec, u(x));
pol := subs({c1=0, c2=0}, rhs(sol));
plot(pol, x=a..b);
dpol := diff(pol, x);
evalf(subs(x=VAL, pol));
evalf(subs(x=VAL, dpol));

```

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dsolve(... = ..., u(x)).
pol := subs({c1=0, ..., cn=0}, rhs(sol))
plot(pol, x=-11..5)
f := unapply(pol, x);
f(0)
f(...)
evalf(f(...))
D(f)(P1/2)
evalf(D(f)(...))

```

4. Find fixed points of map, behavior of the sequence of iterations starting with 1, 0.2, 0.7, 0, ...

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eg := ...
solve(eg, x)
f := unapply(..., x); solve(f(x)=x, x)
f(0) ... (fixed points)

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x := 0
for i from 1 to 100 do
  x := f(x);
  psi(i) := x
od
points := [[m, psi(m)] $ m=1..100];
pointplot(points, symbol=circle)

```

Euler: (approximate solution) $y_{n+1} = y_n + h \cdot f(x_n, y_n)$

with (details):

$f(x, y) \rightarrow 2 \dots$

$dsolve(\{diff(y(x), x) = f(x, y(x)), y(0) = 1\});$

$\phi := unapply(\frac{2}{5}, rhs(\%0), x)$

$DEplot(diff(y(x), x) = f(x, y(x)), y(x), x=0..1.5, [[y(0)=1]], y=1..2.5);$

$h := 0.1$

$x := 0; y := 1$

for i from 1 to 15 do

$y := y + h * f(x, y); psi(i) := y; x := x + h; point(....)$

2. Parametric curve:

$plot([x, y, t=a..b], scaling=constrained);$

3. Check if $e(t)$ can be sol. of $\dot{x} = AX$ planar sys.

$x := \dots$

$y := \dots$

$dx := diff(x, t)$

$dy := diff(y, t)$

$eg1 := dx - (ax + by)$

$eg2 := dy - (cx + dy)$

$expand(eg1)$

$expand(eg2)$

$A := Matrix([[-1], [5, 0]])$

$f := (x, y) \rightarrow \dots$

$solve(\{f1(x, y), f2(x, y)\}, \{x, y\}) \rightarrow \text{equl.}$

$Jm := Jacobian([f1(x, y), f2(x, y)], [x, y])$

$Aeg := subs([x=..., y=...], Jm)$

Eigenvalues(A0)

~~Euler~~

Check diagonalizable:

$A := Matrix(\dots)$

$ev := Eigenvectors(A)[1]$

$P := Eigenvectors(A)[2]$

$D := DiagonalMatrix([ev[1], ev[2], \dots])$

$B := evalm(P \&\# D \&\# P^{-1})$

$evalm(A-B)$ Improved Euler (two slopes)

$f := (x, y) \rightarrow \dots$

$dsolve(\dots)$

$\phi := unapply(\dots)$

$DEplot(\dots)$

for ...

$y := y + h/2 * f(x, y) + h/2 * f(x+h, y+h * f(x, y))$

$y := y + h * f(x, y)$

DS Lab Test prep - Maple

restart;

Packages:

with(plots):
 with(DEtools):
 with(LinearAlgebra):
 with(linalg):
 with(VectorCalculus):

Others:

Pi → value
 pi → symbol
 sqrt(2)
 infinity
 I

Differential equations:

ec := diff(x(t), t) + x(t) = 15 } gen sol.
 dsolve(ec, x(t));
 ec := diff(x(t), t^2) + x(t) = 0;
 incond := x(Pi/2) = 1, D(x)(Pi/2) = -2; } IVP.
 sol := dsolve({ec, incond}, x(t));
 phi := unapply(rhs(sol), t);
 phi(t)
 plot(phi(t), t = -4*Pi .. 4*Pi); } plot, limits;
 limit(phi(t), t = infinity)

Test if a function solves an ODE: $\text{odetest}(x(t) = \text{sim}(t), \text{diff}(x(t), t^4) - x(t) = 0)$

every line ends with ; or :

Evaluate numbers: evalf → evaluates num using floating-point precision
 evalf(Pi);
 evalf(sqrt(2));
 evalf(Pi, 20); eval → keeps symbols

Define expressions vs functions:

g := exp(x) - sin(x)
 diff(g, x)
 eval(g, x = 0)
 int(g, x)

Functions

f := x → exp(x) - sin(x) → define a function
 f(0)
 D(f)(0) D → for functions
 D(f)(x) diff → for variables
 (D@@2)(f)(x)

f := unapply(exp(2x) - cos(2x), x) → define a function

Algebraic equations:

solve(x^2 - 4x + 3 = 0, x); → might not calculate correctly (especially for trigon.)
 fsolve(x - cos(x) = 0, x); → approximate sol.
 → if solve does not work, try fsolve

solve({eq1, eq2}, {x, y}) → solve system.

Linear Systems:

sys := {diff(x(t), t) = ..., diff(y(t), t) = ...};
 incond := {x(0) = 1, y(0) = 1};
 sol := dsolve(sys union ic);

Matrix stuff: with(LinearAlgebra)

A := Matrix([[-1, -3], [3, -1]]); ⇒ $\begin{bmatrix} -1 & -3 \\ 3 & -1 \end{bmatrix}$
 Determinant(A)
 Eigenvalues(A)
 Eigenvectors(A)
 CharacteristicPolynomial(A, lambda)
 MatrixExponential(A)
 A⁻¹

Plot the planar curve of param. equations:

plot([x-expression, y-expression, t = a..b],
 scaling = constrained);

Hyperbolic equilibrium → All eigenvalues have $\text{Re}(\lambda) \neq 0$